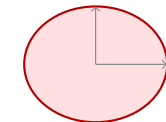


Manipulability: how dexterous is a posture?

$$w = \sqrt{\det(JJ^T)} \quad [\text{Yoshikawa 1985}]$$

- J — the arm's **Jacobian** at configuration q , mapping joint rates to tool velocity: $\dot{x} = J\dot{q}$ (here 6×8 : 6-DOF pose, 8-DOF chain).
- JJ^T collapses the rectangular J into a 6×6 matrix whose determinant we can take.
- w is the **volume of the manipulability ellipsoid** — the tool velocities reachable from a unit sphere of joint rates ($w = \sigma_1\sigma_2 \cdots \sigma_6$, the singular values of J).

■ **Large w :** fat, round ellipsoid — moves easily in any direction (dexterous).



high w (dexterous)



low $w \rightarrow 0$ (near singular)

Manipulability ellipsoid: axes = singular values of J .